

Stefan Rogers-Coltman

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Neuroscientist with a PhD from the MRC Laboratory of Molecular Biology, bringing strong practical experience in end-to-end design and development. Skilled in combining scientific domain knowledge with hardware, electronics, and software engineering to deliver reliable, integrated systems. Experienced collaborating with specialists across disciplines, with a track record of rapidly engaging with unfamiliar technical domains.

Education

PhD, Systems Neuroscience 2021–2025

University of Cambridge, Cambridge, UK

Supervisor: Dr. Marco Tripodi, MRC Laboratory of Molecular Biology.

MSc, Translational Neuroscience 2020–2021

Imperial College London, London, UK

BSc (Hons), Biomedical Sciences 2017–2020

University College London, London, UK

1st Class Honours. Neuroscience specialisation.

Research Experience

Postdoctoral Researcher 2025–present

MRC Laboratory of Molecular Biology, Cambridge, UK

Extending the behavioural platform developed during my PhD, working across the scientist-engineer interface and supervising a computing student.

- Led development of a fully wireless successor to the behavioural rig system detailed below, taking it from concept to manufactured hardware in weeks: designed custom PCBs and wrote MCU firmware (STM32, BLE)
- Designed and built a mouse-wearable wired motion sensor (9-DOF IMU with magnetometer)

Doctoral Researcher 2021–2025

MRC Laboratory of Molecular Biology, Cambridge, UK

Initiated and led a cross-functional research programme, developing a scalable experimental platform integrating hardware, software, and neuroscience methods, now adopted across multiple projects within the lab. Collaborated closely with researchers and mechanical/electrical workshops to design, build, and deploy robust experimental systems.

Experimental Systems & Engineering

- Designed and built automated behavioural rigs for studying spatial decision-making in freely moving mice
- Developed wireless load-cell platforms for automated trial initiation
- Implemented high-speed video acquisition systems in C++ using Spinnaker API, enabling synchronised multi-camera recording
- Engineered general-purpose data acquisition architecture for synchronising behavioural, video, and electrophysiology signals, with end-to-end pipelines for storage and analysis
- Built integrated optogenetics control systems combining IMU-based head tracking, automated laser stimulation, and real-time data visualisation

Systems Neuroscience & Circuit Mechanisms

- Developed a novel behavioural paradigm dissociating sensory input from motor output in a naturalistic setting
- Manipulated superior colliculus Pitx2+ neurons using chemogenetics (DREADDs), demonstrating modulation of baseline spatial priors across movements
- Performed chronic electrophysiological recordings (optetrodes, Cambridge NeuroTech, Neuropixels) in freely moving mice during behaviour
- Identified a subpopulation of neurons encoding spatial confidence, selectively active during correct visual decisions but not during incorrect or non-visual trials
- Integrated neural and behavioural datasets to link circuit-level activity with decision-making processes

Selected Additional Experience

Teaching Assistant 2026

Cambridge Open Lab: Open Science in Practice Summer School,

Taught engineering techniques for neuroscience to PhD students, assisting practical workshops in electronics design (Arduino) and CAD (FreeCAD).

Head of Maintenance (Bosun) 2026

Cambridge University Yacht Club,

Responsible for maintenance and operational readiness of club yacht. Diagnosing and resolving mechanical and electrical issues to ensure safe and continuous operation.

MSc & BSc Research Projects 2019–21

Imperial College London / UCL, London, UK

Developed experimental systems for generating senescent cell cultures (MSc); investigated noradrenergic modulation of Golgi cell inhibition of cerebellar granule cells (BSc).

Founder 2020

FPV Drone Build Venture,

Designed and built custom FPV drones; managed end-to-end assembly and client delivery.

Research Assistant Summers 2018/19

MRC LMB / UCL, Cambridge & London, UK

- Developed organoid culture techniques (inner ear and choroid plexus models)
- Performed imaging and analysis; contributed to protocol optimisation and automated sampling approaches

Skills

Programming & Software	Python (data analysis, automation), C++ (high-performance imaging), embedded firmware (STM32, BLE, Arduino)
Experimental Systems & Engineering	Behavioural rig design, CAD and PCB design (Solidworks, KiCad), wireless sensor systems, high-speed imaging (FLIR / Spinnaker), hardware–software integration, multi-system synchronisation
Neuroscience & <i>In Vivo</i> Methods	UK Home Office Personal Licence (PIL) holder; behavioural task design and animal training; mouse neurosurgery (viral injections, implants); chronic electrophysiology (optetrodes, Neuropixels)
Data Analysis	Statistical modelling, custom pipeline development, neural data integration

Publications

Manuscript in preparation — behavioural paradigm and superior colliculus manipulation study (Rogers-Coltman S, Welch D, Geyer L, Tripodi M.).

Awards & Funding

Laidlaw Undergraduate Research and Leadership Scholarship

Laidlaw Foundation

Funded research placements and leadership training